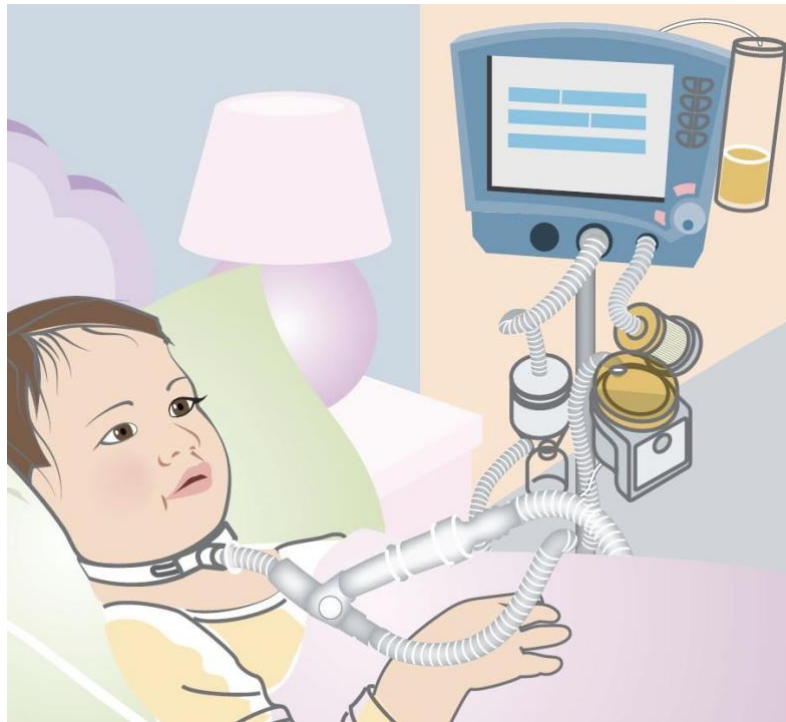


Pediatric Home Ventilation Team

Structure, Function and Guidelines

For Safe Home Transition and Better Home Care



Pediatric Home Ventilation Team
The Royal Hospital
Child Health Department

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Preface

Dr. Saif Awlad Thani
Pediatric intensivist

The Pediatric Home-Ventilation team is a multidisciplinary group officially established within the Child Health Department in May 2018. Its primary purpose is to facilitate the transition and ongoing care of children requiring tracheostomy and mechanical ventilation at home. Given the complexity of these cases, reliance on specialized equipment, the need for comprehensive family education, regular follow-up, and limited community health support, a diverse team is essential for effective care provision. Comprising physicians, nurses, respiratory therapists, discharge planners, dietitians, social workers, physiotherapists, occupational therapists, and speech pathologists, each member contributes according to their expertise to ensure optimal care delivery. The team's responsibilities extend from hospital admission to home care and follow-up visits. Paramount among these duties is ensuring the safe transition of patients to their homes and equipping families with the necessary knowledge and skills to care for their child in a home environment. To support these objectives, we have developed a comprehensive booklet covering all practical aspects of pediatric home-ventilation care. This includes detailed descriptions of team member roles and responsibilities, as well as a home-transition pathway outlining every step of the process. Our aspiration is to lead in pediatric home-ventilation care in Oman and share our experiences with other hospitals in the country.

Vision and mission

Mission:

Our mission is to deliver multidisciplinary, high-quality care to children necessitating invasive home ventilation via tracheostomy. We wholeheartedly endorse the medical home model. Our objective is to ensure the safe and smooth transition of children from hospital to home, with diligent follow-up, reduced hospital stays, and enhanced quality of life for both children and their families.

Vision:

Our vision is to spearhead a nationwide pediatric home ventilation program, setting the standard for superior care for children reliant on home ventilation across the entire country

Roles and Responsibilities

- **Physician:**

- 1) Receives referrals to the team, assesses patients, and takes the lead in medical management, including ventilation adjustments, decannulation, and other necessary interventions.
- 2) Provides essential emergency training to families, covering Basic Life Support techniques and tracheostomy reinsertion procedures.
- 3) Completes home equipment forms and submits them to the social worker for further processing.
- 4) Reviews the patient discharge checklist alongside the designated team member, ensuring that all criteria are met before discharge.
- 5) Develops a comprehensive follow-up plan, specifying the schedule for tracheostomy changes, daycare visits, and equipment inspections.
- 6) Conducts regular follow-up assessments of patients in daycare settings, monitoring their progress and adjusting care plans as needed

- **Discharge planner:**

- 1) Coordinate multidisciplinary team (MDT) meetings upon patient referral to the home ventilation team, scheduling subsequent meetings until patient discharge.
- 2) Take charge of MDT meetings, establishing agendas, determining discharge dates in collaboration with core MDT members and the family
- 3) Monitor progress along the home transition pathway by maintaining communication with other home ventilation team members, ensuring readiness for discharge.
- 4) Provide ongoing support to the family, regularly reassessing their needs and issuing supportive letters utilizing prepared templates for home ventilated patients
- 5) Post-discharge, arrange daycare appointments based on physician assessments and patient requirements. The initial daycare appointment is scheduled one month after discharge, followed by appointments every three months unless otherwise directed by the physician
- 6) Coordinate booked admissions for re-evaluation and manage transfers to other hospitals or home as necessary

- **Nurse:**

- 1) Educates the family on tracheostomy care (bagging, suctioning, hygiene, and tie changes) utilizing the family teaching checklist
- 2) Instructs parents on administering NGT or gastrostomy feeds and teaches them how to safely insert NGT tubes at home
- 3) Ensures that the family is proficient in operating the home suction machine prior to home transition
- 4) Offers ongoing support to the family, fostering confidence and proficiency in caring for their child at home
- 5) Conducts follow-up phone calls with the family on the evening of discharge, after 24 hours, and after 72 hours to monitor progress and address any concerns
- 6) Assists in teaching parents how to manage emergency situations in the absence of a physician, ensuring they are prepared to handle critical scenarios effectively

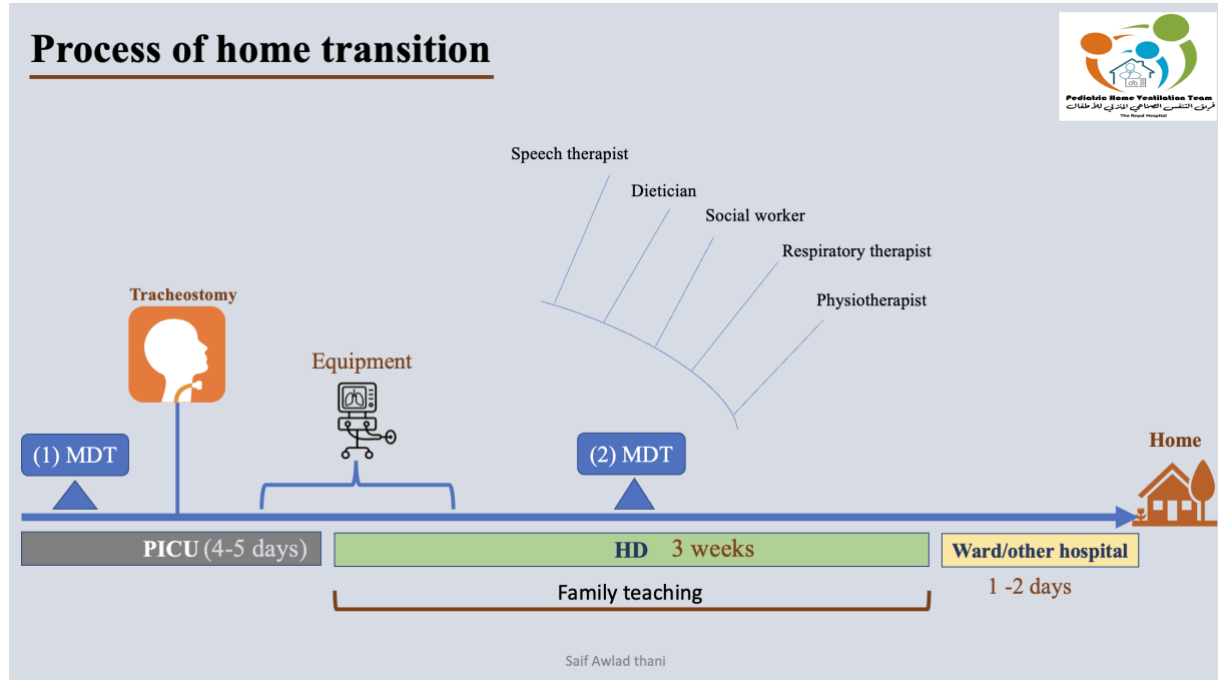
- **Speech pathologist:**

- 1) Conducts oro-motor and swallowing assessments either at the bedside or instrumentally to determine the most suitable feeding route.
- 2) Following consultation and collaboration with other team members, conducts a comprehensive assessment to evaluate the feasibility of connecting a speech valve through the ventilator for communication and voice projection. Discusses the necessity for consultations with other teams to facilitate successful restoration of communication.
- 3) Assists with tracheostomy changes and provides specialized care if a speech valve is in place.
- 4) Contributes to the education of family members and other caregivers regarding the impact of tracheostomy on speech and swallowing, as well as the potential for restoring these functions with the tracheostomy and ventilator in place.
- 5) Participates alongside other team members in family meetings and aids in preparing both the family and patient for the transition to home care
- 6) Ensures proper follow-up care is provided in coordination with other team members.

- **Respiratory Therapist:**
 - 1) Collaborates with the Ministry's home care service, other team members, and the family to ensure the availability of all necessary home care equipment prior to the transition home, tailored to the patient's specific needs. (Details of required equipment are outlined in the home equipment section.)
 - 2) Educates the family on the proper usage and maintenance of home ventilation equipment, including the ventilator, humidifier, circuit connections, oxygen concentrator, nebulizer, and oxygen cylinder.
 - 3) Ensures timely maintenance of home care machines to prevent unexpected malfunctions.
 - 4) Conducts checks on the home care machine during patient follow-up appointments.
 - 5) Provides the family with any required accessories associated with the home care equipment to ensure optimal functionality and patient comfort.
- **Physiotherapist:**
 - 1) Implements early mobilization protocols and conducts regular physiotherapy sessions to prevent muscle weakness and atrophy in patients.
 - 2) Provides comprehensive instruction to the family on performing chest physiotherapy at home and proper suction techniques.
 - 3) Educates the family on proper positioning techniques to optimize comfort and support for the patient.
- **Occupational therapist:**
 - 1) Ensures the safety and accessibility of the home environment for both the child and the family.
 - 2) Educates the family on methods to enhance the quality of life for both the child and the family, focusing on improving the child's functional performance and exploring adaptation options.
 - 3) Conducts assessments and instructs the family on techniques to enhance handling, positioning, joint flexibility, and motor skills for the child.
 - 4) Assesses and implements strategies to improve the child's oral motor skills to prevent sensory issues and promote optimal development.
- **Dietician:**
 - 1) Conducts a thorough assessment of the patient's nutritional status.
 - 2) Offers dietary guidance to ensure the patient receives sufficient calories and meets their optimal nutritional requirements.
 - 3) Assists in determining the most appropriate feeding route for the patient.
 - 4) Arranges for the provision of a feeding pump for patients requiring continuous feed upon discharge.
- **Social Worker:**
 - 1) Submits equipment request forms to the Respiratory Care Services within the Ministry of Health and diligently follows up on the availability of the requested equipment.
 - 2) Assists the patient's family in obtaining necessary equipment from various sources, such as the Ministry of Health, hospitals, charities, companies, and family members, taking into account the family's financial status.
 - 3) Engages in communication with the patient's parents and close family members to gather psychological, social, and financial details crucial for comprehensive care planning.
 - 4) Provides guidance and support to prepare the family for accepting and adapting to the patient's new medical situation.
 - 5) Ensures that parents can create an environment that meets the patient's needs and guarantees their safety.
 - 6) Collaborates with the Ministry of Social and Development Affairs to secure appropriate financial and social support for the patient as needed.

Clinical pathway (Tracheostomy to Home transition)

Dr. Saif Awlad Thani
Pediatric intensivist



This clinical pathway outlines the progression from the decision to undergo tracheostomy to the transition home. The duration of each step may vary depending on the patient's needs, equipment availability, and family readiness. Upon the decision for tracheostomy by the primary physician, the patient is referred to the home ventilation team, except in cases of emergency tracheostomy.

The initial multidisciplinary team (MDT) meeting occurs prior to tracheostomy insertion. Following the procedure, patients typically spend approximately 4-5 days in the pediatric or post-cardiac intensive care unit, unless extended stay is warranted for additional medical concerns at the discretion of the treating physician. Neonatal patients with tracheostomies remain in the neonatal intensive care unit until they are ready for home transition.

Equipment preparation commences promptly after the first MDT meeting. Family education begins in the intensive care unit but primarily takes place in the high dependency unit once the patient's condition stabilizes. A second MDT meeting is convened after approximately two weeks in the high dependency unit to assess teaching progress, family readiness, equipment readiness, and determine the home transition date.

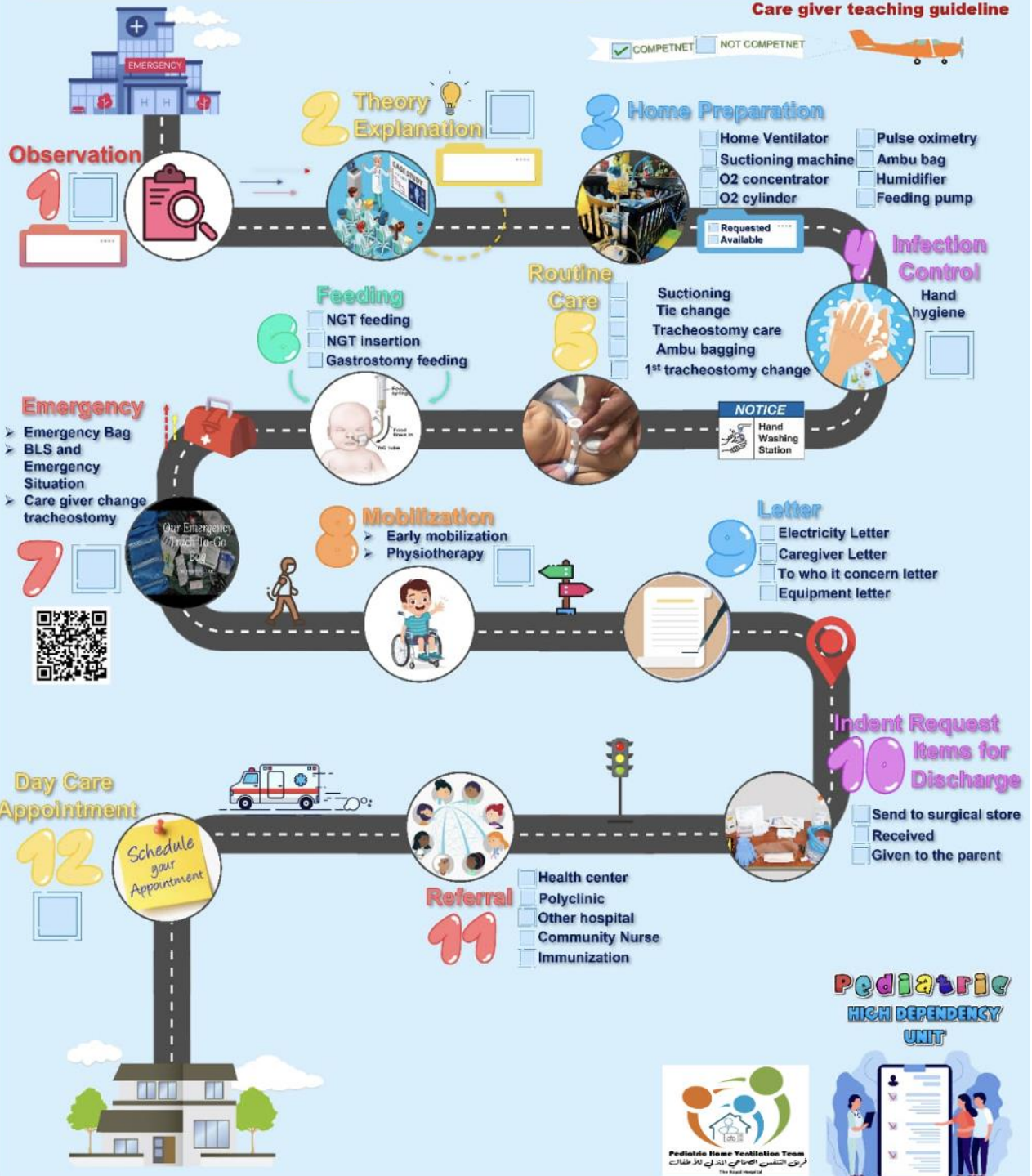
Upon readiness for discharge, patients may be transferred to a regional hospital for one or two days or moved to the general ward, where full care responsibilities transition to the family (refer to the home simulation day section).

YOUR PATHWAY TO HOME

Tracheostomy



Care giver teaching guideline

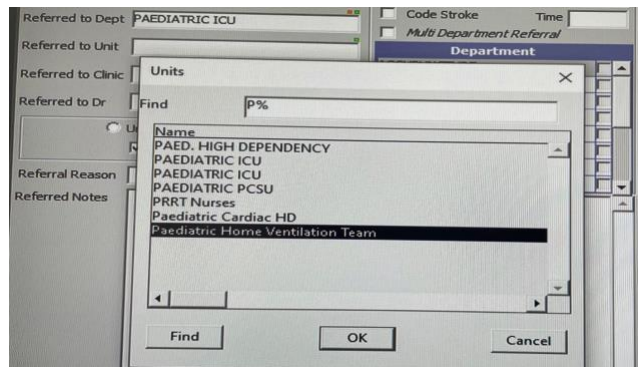


How to activate/refer to pediatric home ventilation team?

Dr. Said Al Hanshi
Pediatric intensivist

Referral of new patient:

For new patients requiring tracheostomy tube insertion, an immediate referral to the Pediatric Home ventilation team should be made. This process allows for counseling of parents or caregivers regarding home ventilation process and initiates the preparation of necessary equipment and educational plan. Depending on the patient's needs, they may require either tracheostomy alone or both tracheostomy and home ventilation. There may be instances where uncertainty arises regarding the necessity of home ventilation, particularly concerning the patient's ability to be weaned off mechanical ventilation. In all cases, early referral to the Pediatric Home Ventilation Team is essential for comprehensive assessment and appropriate planning. In “Al Shifa” system, go to consults and choose: Referred to Dep: Pediatric ICU then Referred to unit: Paediatric Home Ventilation Team, as following:



- In the referral, please include the following information:
 1. Diagnosis/problems list
 2. Reason for tracheostomy/home ventilation
 3. Proposed date of tracheostomy insertion
 4. Confirmation of family awareness about tracheostomy decision
 5. Indication of whether the patient was evaluated by ENT or pulmonology team

Following the referral, contact the physician covering the home ventilation team, using the schedule provided in the department's rota. The physician will review the patient and coordinate the first multidisciplinary meeting in collaboration with the patient's primary physician. During the initial meeting, the following home ventilation team members should be present: discharge planner, physician, respiratory therapist, social worker, and any other relevant team members based on the patient's condition.

The purpose of this meeting is to:

- Introduce the home ventilation team to the parents/caregivers
- Explain the home transition pathway, including teaching, equipment, and support provided by the home ventilation team
- Discuss the follow-up plan after discharge
- Address any questions or concerns from the parents/caregivers.

Referral of known patient:

If the patient is already tracheostomized and known to home ventilation team, and is admitted to PICU, High Dependency Unit, or pediatric wards (Pediatric, Pediatric Surgery, Pediatric Oncology, Pediatric Cardiology), the referral process to the Pediatric Home Ventilation Team can be facilitated through the following steps:

1. Initiate an internal referral in the Al-Shifaa System, including pertinent patient information such as current diagnosis, reason for referral, and any relevant medical history. Please note that pediatric home ventilation team does not treat acute condition, but it follows up any concern regarding home ventilation.

2. Call the physician responsible for covering the Pediatric Home Ventilation Team as per the provided schedule.
3. The covering physician will review the patient's status and coordinate any necessary actions with the home ventilation team.

Referring a new patient, who is not previously known to our team but is tracheostomized and discharged from another hospital while on home ventilation:

If a child who has been tracheostomized and placed on home ventilation from another hospital is admitted for acute issues, procedures, or evaluations, the following steps must be taken within 24 hours:

- 1) Referral to Pediatric Home Ventilation Physician and Respiratory Therapist: The patient must be referred within 24 hours of admission to the pediatric home ventilation team and respiratory therapist for overall assessment.
- 2) Equipment Check by Respiratory Therapist: The respiratory therapist will inspect the home ventilator, circuits, and humidifier to ensure proper functioning and activation of alarms.
- 3) Comprehensive Assessment by Home Ventilation Physician: The home ventilation physician will conduct a thorough review of the patient's condition, including assessment of the tracheostomy, ventilator settings, and reevaluation of oxygenation and ventilation requirements.
- 4) Family Education Review: The physician will reassess the family's competency in tracheostomy care, basic life support, and management of tracheostomy emergencies to ensure adequate understanding and readiness for home care.

Family teaching

Asma Al Naabi
Home Ventilation Nurse

Family teaching is paramount in preparing a tracheostomized and ventilated child for a safe transition home. As community services for home ventilated children are currently unavailable, full care at home falls upon the parents. Ensuring the family's competence and confidence in caring for their child at home is crucial before the transition. They are trained to recognize signs that their child is unwell, including changes in color, level of consciousness, and respiratory effort, emphasizing the importance of not relying solely on pulse oximetry or ventilator alarms. Although many parents may initially feel apprehensive about learning tracheostomy care, structured teaching helps them gain confidence over time. Family teaching is divided into three domains:

1. Tracheostomy care
 2. Equipment care
 3. Emergency situation
- **Tracheostomy care:**
Mastering tracheostomy care is paramount for families before their child can safely return home. Teaching commences in the intensive care unit and intensifies during their stay in the high dependency unit. It is imperative that at least two caregivers become proficient in all aspects of tracheostomy care, and we strongly advocate for a third caregiver to also undergo training. This ensures continuity of care at home, even when one or two caregivers are unavailable. Families are educated on the following key points:
 - 1) Tracheostomy skin care
 - 2) Tracheostomy suctioning
 - 3) Oral and nasal suctioning
 - 4) Tracheostomy tie change
 - 5) Oral care
 - 6) Tracheostomy tube change
 - 7) Feeding
 - 8) Medication administration

In English and Arabic, we designed a booklet explaining everything a family needs to know about caring for a child with tracheostomy and home ventilation. It can be downloaded via this QR code.



A competency checklist is attached to ensure all steps are taught and understood. It should be signed by the healthcare worker providing the teaching and by two family members to confirm their understanding and competency.

The **respiratory therapist** is responsible for educating the family about home equipment, which includes the ventilator, suction machine, oxygen concentrator, nebulizer, and oxygen cylinder. The respiratory therapist teaching checklist outlines all the points covered by the Respiratory Therapist before the transition home. Multiple sessions may be conducted for the family as needed.



The Pediatric Home Ventilation Team utilizes three manikins for family teaching purposes.

المستشفى السلطاني فريق الأطفال للتنفس الصناعي المنزلي استمارة تدريب

اسم الطفل.....
قبل ذهاب طفلك إلى المنزل يتعين عليك وعلى مقدم الرعاية المساعدة اكمال الاستمارة التالية التي توضح العناصر التي عليك تعلمها قبل انتقال طفلك للمنزل.
عند اكمالك محاولة ناجحة قم بوضع توقيعك على كل عنصر تم تدريبك عليه وعندك اكمالك 3 محاولات ناجحة يقوم الممرض بوضع توقيعك على العنصر الذي تم استكمال تدريبك فيه.

عناصر التدريب	مقدم الرعاية / care giver	توقيع الممرض Nurse sign	مقدم الرعاية / care giver	توقيع الممرض Nurse sign
1	طريقة الصحيحة لتعقيم اليدين / Hand hygiene			
الاهتمام بأنبوب التنفس العنقي				
2	نوع أنبوب التنفس العنقي / Type of tracheostomy			
3	قياس أنبوب التنفس العنقي / Size of tracheostomy			
4	كيفية تحضير ربطة التنفس العنقي / How to prepare trach tie			
5	متى يتم تغيير ربطة أنبوب التنفس العنقي / Time to change trach tie			
6	تغيير رباط أنبوب التنفس العنقي / Trach. tie change			
7	تنظيف أنبوب التنفس العنقي / Tracheostomy cleaning			
8	موعد تغيير أنبوب التنفس العنقي / Time to change trach.			
9	العلامات الدالة على وجود التهاب / Signs of infection			
شفط البلغم				
10	اللون الطبيعي للبلغم / Normal secretion color			
11	القياس المناسب لأنبوب الشفط / size of suction tube			
12	الطول المناسب لأنبوب الشفط / Length of suction tube			
13	طريقة استخدام بالون التنفس الاصطناعي / Bagging			
14	الضغط المناسب لشفط البلغم / Suction machine pressure			
15	شفط البلغم من أنبوب التنفس العنقي / Tracheostomy suction			
16	شفط البلغم من الفم والالنف / Nasal oral suction			
17	المدة الزمنية المناسبة لشفط البلغم / Duration for trach suction			
18	تنظيف أنبوب الشفط / Cleaning suction tube			
19	كل كم يتم تغيير أنبوب الشفط / How often Suction tube to be changed			
التغذية				
20	التغذية بأنبوب المعدة / NGT feed			
21	أدخال أنبوب التغذية / Insert NGT			
22	التغذية بالأنبوب المعوي / Gastrostomy Feed			
العناية اليومية				
23	الاستحمام / Bath			
24	تنظيف الفم / Oral care			
25	إعطاء الأدوية / Medication administration			
26	تنظيف المعدات الطبية / Equipment cleaning			
حالات الطوارئ				
27	حقبة الطوارئ / Emergency bag			
28	الإنعاش وحالات الطوارئ / BLS & emergency situations			

لقد قرأنا وفهمنا وتدريبنا بما يجب علينا تعلمه لتوفير رعاية لطفلا قبل خروج الطفل من المستشفى وانتقاله للمنزل. كما تم الإجابة على أسألتنا المتعلقة بكيفية الاهتمام بأنبوب التنفس العنقي.
الاسم الكامل لمقدم الرعاية الاسم الكامل لمقدم الرعاية المساعدة
توقيع مقدم الرعاية توقيع مقدم الرعاية المساعدة

Pediatric Home Ventilation Team The Royal Hospital

Respiratory therapist teaching checklist

(Adopted and modified from "Home Care Assessment form" by

Respiratory Care Services, MOH)*

No	Assessment	Date	RT name
1	Ventilator - How to turn on/ Off the machine - How to put the machine on standby mode - Importance of Services and maintenance of the machine - When is the machine due for service - How to clean the machine - Explain briefly about the current ventilator mode patient on and not to change without doctor's order. (Put on lock mode if this feature is available) - Explain about the battery backup duration of the ventilator. Keep the ventilator on charge all the time. - Make sure that the relative understands the need for ventilator accessories and the duration of change (HME & catheter mount, 2times a week unless it is dirty)		
2	Circuit: - Show how to change the ventilator circuit and how to connect new circuit. (Changes once in a month unless it is dirty) - Explain about the importance of expiratory valve and should not be occluded		
3	Battery Backup: - Ventilator must have a backup battery life of 4 to 8 hrs. That is most important during transport. - Portable O2 Concentrator - Car Charger as alternative - Availability of External Battery backup if electricity was suddenly disconnected		
4	Basic Troubleshooting - Alarms Setting: - Important attention to all alarms - Differentiate between normal alarm & emergency alarm - Low Pressure Alarm ex. Circuit Leak - High Pressure Alarm because of secretions, coughing, etc. - Disconnection alarm		
5	Humidification: - Explain the importance of using humidifier - Which type of humidification to use (Active/HME) - What type of water to use Importance of Cleaning the humidifier		
6	The environment - The place should be clean & well-ventilated area - Should not be congested - Electrical Socket must be more than (1) one to prevent error from happening - O2 concentrator should not be near to window to prevent accumulation of dust		
7	Oxygen: - Instruct the relative about the use of Oxygen source and connection (Cylinder / Concentrator). - Explain how to connect Oxygen concentrator and Oxygen Cylinder to the ventilator and to Ambu bag		

"Home Care Assessment form" by Respiratory Care Services, MOH



- **Emergency situation teaching:**

Caring for a home ventilated child dependent on equipment presents significant challenges, particularly when faced with emergency situations. Dealing with such events at home can be daunting and anxiety-inducing for families. Therefore, it is essential to provide comprehensive training on how to manage unexpected emergencies. These include tracheostomy blockage, tracheostomy dislodgement, ventilator malfunction, and respiratory or cardiac arrest. Families are also instructed in Basic Life Support techniques, with adjustments tailored to the needs of a tracheostomized child.

Emergency situation training typically spans multiple sessions, allowing families ample opportunity to master the skills necessary for effective response. It is imperative that families feel competent and confident before their child is discharged home. As part of this preparation, at least one family member must demonstrate proficiency in changing the tracheostomy tube under supervision. This practice is conducted both on a manikin and on the child to ensure readiness for real-life situations.

Preparation of home equipment

RT. Khalsa Al Syiabi
Head of Respiratory services, MOH

After the patient has been determined to require a tracheostomy and home ventilation, and the initial multidisciplinary meeting has been convened, equipment preparation becomes a priority. The home ventilation physician is responsible for determining the necessary equipment for the patient, while the team reviews the available equipment within the hospital. Simultaneously, the social worker evaluates the financial status of the family to determine their ability to procure some equipment independently.

Following the social worker's assessment and collaboration with the family, the physician completes an equipment request form and submits it to the social worker. The social worker then communicates with the Directorate General of Specialized Medical Care under the Department of Supportive Services via Barwa's email system. They continue to monitor the progress until the patient receives their equipment.

The essential home equipment may comprise:

1. Home ventilation machine with humidifier
2. Suction machine
3. Oxygen concentrator (if the patient requires oxygen)
4. Pulse oximeter with alarm function
5. Backup oxygen cylinder (even if oxygen is not currently needed)
6. Ambu bag appropriate for the patient's age and weight
7. Nebulizer machine (if the patient requires nebulized medication)

Once the equipment is provided, the family is required to sign an undertaking form (attached below) confirming receipt of the equipment and agreeing to return it to the hospital once it is no longer needed.

وزارة الصحة
المديرية العامة للرعاية الطبية التخصصية
قسم الرعاية التنفسية
استمارة تعهد

NO :

التاريخ:

الموافق:

المستشفى :

- Ventilator
- O2 Concentrator
- NIV (BIPAP- Auto CPAP)
- Manual Resuscitator
- Humidifier for Pediatrics Population
- Pulse Oximetry

أنا الموقع أدناه استلمت جهاز

لاستخدامه للمريض: رقم الملف الطبي (.....) . نوع القراية ()
لذا أتعهد بحفظه وإعادةه إلى قسم الرعاية التنفسية في حالة تحسن صحة المريض ولا يحق لي التصرف فيه أو
إعطائه لشخص آخر وإلا سأتحمل المسؤولية.

اسم المستلم :

رقم الهاتف :

التوقيع :

سلم بواسطة الأخصائي التنفسي:

التوقيع :

رئيس قسم الرعاية التنفسية

نسخة إلى :

- صاحب العلاقة
- يحفظ في ملف قسم الرعاية التنفسية .

Sultanate of Oman
Ministry of Health
Directorate General of Specialized Medical Care
Medical Prescription (Respiratory Care Equipment for Home-use)

وصفه طبيه (اجهزة الرعاية التنفسية الخاصة للاستعمال المنزلي)

To: The Head of the Respiratory Care Services, Directorate General of Specialized Medical Care

Hospital: _____ Date: _____	
Patient Phone: _____ Patient Phone / GSM #: _____ Patient home address: _____ Civil ID Number (must): _____	Patient Sticker (Sticker only Hand writing is not acceptable) Body Weight: _____ Age: _____
هام جدا: جميع اجهزة التنفس الخاصة للاستعمال المنزلي هي ملك لوزارة الصحة؛ وبناء عليه يجب اعادتها فور الانتهاء من استعمالها	
<u>Respiratory Diagnosis, select [x] from the following:</u> ☐ COPD ☐ OSA ☐ CRF II ☐ Obesity Hypo-ventilation ☐ Other, indicate: _____	
<u>Patient's Current Situation Select [x]:</u> Select one: ☐ Patient is not intubated ☐ Intubated with ET-Tube ☐ Intubated w/ Tracheostomy-Tube Select one: ☐ Patient is using hospital equipment ☐ Using donated equipment ☐ Using condemned in-use equipment Select one: ☐ Patient is available at the hospital ☐ Patient is available at home ☐ Other, indicate _____	
<u>Non-Invasive Ventilation Equipment, Select [x] one</u> ☐ Auto CPAP alone Setting RR () – IPAP () – CPAP or EPAP () ☐ BIPAP S / ST alone ☐ BIPAP w/ O2 Concentrator 1-5 L/M ☐ BIPAP w/ O2 Concentrator 5-10 L/M	
<u>Portable Ventilator/Brand ()</u> Select one: ☐ Portable ventilator with adult circuit. ☐ Portable ventilator with child/infant circuit. <u>Ventilator Accessory</u> Ventilator Setting RR () – PIP () – PEEP () – PS () – O2 () Select one: ☐ Oxygen Concentrator 1-5 L/M ☐ Oxygen Concentrator 5-10 L/M Select one: ☐ Manual Suction Pump ☐ Electric Suction Pump Select one: ☐ Adult Manual Resuscitator ☐ Child Manual Resuscitator ☐ Neonate Manual Resuscitator	
<u>Department Head/ Consultant</u> Name (print): _____ Signature: _____ Date: _____ Stamp: _____	<u>Respiratory Therapist</u> Name (print): _____ Signature: _____ Date: _____ Stamp: _____
cc: ☐ Respiratory Care Services ☐ Discharge Planner ☐ Social Worker	

Tracheostomy Tube change

Dr. Saif Awlad Thani
Pediatric Intensivist

- **First Change:**

The first tracheostomy tube change is typically performed once the tracheostomy stoma has matured, usually after one to two weeks of tracheostomy tube placement. However, it may be necessary to perform the change earlier for upsizing or downsizing the tracheostomy tube if clinically indicated. This initial change is conducted by a pediatric surgeon, who also removes the stay suture during the procedure. It can be performed in various settings, including the intensive care unit, high dependency unit, or ward, provided that emergency airway equipment and personnel support are readily available. In cases where the tracheostomy tube change is required earlier than one-week post-placement, the surgeon must assess whether the patient should be taken to the operating theater due to the increased risk of insertion difficulty resulting from the non-maturation of the stoma. The location of the first tracheostomy change should be determined by the pediatric surgeon. However, in the event of an airway emergency such as tube dislodgement or complete blockage, the change can be performed anywhere without delay if there is insufficient time to transfer the patient to a critical care unit or operating theater.

As part of the preparation for home transition, it is mandatory for at least one parent to perform the reinsertion of the tracheostomy tube in their child. There is no requirement to open a new tracheostomy tube for this purpose unless it is due for scheduled replacement; instead, the same tracheostomy tube can be removed and reinserted.

- **Routine change:**

The routine change of the tracheostomy tube is typically performed every 30 to 90 days, or as per the manufacturer's recommendations based on the type and material of the tube.

Our goal is to empower parents to perform the tube change initially under supervision, particularly during daycare visits, especially within the first year after transitioning home. As they gain confidence and competency, and upon approval by a physician from the pediatric home ventilation team, they can eventually perform the tube change independently, even at home.

Please refer to the appendix for detailed instructions on how to prepare for and perform the tracheostomy tube change.

Equipment maintenance

Saif Al Balushi
Respiratory therapist

Home ventilation devices require regular checks and maintenance to ensure optimal functionality. During daycare visits every 3-6 months, or during hospital admissions for any reason, the home ventilator undergoes a bedside check by a respiratory therapist to confirm its proper operation. Additionally, every 6 months, a comprehensive check and maintenance of the ventilator should be conducted. If the ventilator was provided and registered in the hospital, this maintenance is performed by the hospital biomedical department. However, if the ventilator was purchased by the parent or provided by the home care service (MOH), maintenance is carried out by the respective companies.

Families are encouraged to bring any other equipment they have issues with during these visits, so that they can be assessed and addressed accordingly. This proactive approach ensures the continued reliability and safety of all home ventilation equipment.

Equipment supply procedure

Dr. Saif Awlad Thani
Pediatric Intensivist
&
Asma Al Nabi
Home ventilation nurse

Patients who are equipment-dependent require a range of accessories to be supplied at home, tailored to their specific needs. These may include suction catheters, HME filters, ventilator circuits, tracheostomy cleaning kits, NGT (Nasogastric Tube), gloves, and other necessary items. Physicians and nurses collaborate to determine the appropriate accessories for each patient.

Upon discharge from the hospital, the patient receives a supply of accessories for a duration of 3 months, provided by the Royal Hospital. Additionally, a referral is issued to the local health center or regional hospital,

where the patient or their caregiver can regularly collect these supplies. The discharge planner may be required to liaise with other healthcare centers or hospitals to facilitate the supply of accessories before discharge, or address any challenges encountered by families later on.

It is crucial for the patient to have spare tracheostomy tubes at home, ensuring preparedness for emergencies. They should possess one tube of the same size as their current tube, as well as one tube of a smaller size. For example, if the patient's tracheostomy tube size is 4.0, they should have both a spare tube of size 4.0 and a tube of size 3.5. Families are instructed that in the event of a tracheostomy dislodgement at home, if they are unable to reinsert the same size tube, they should attempt to insert the smaller size tube. This preparation enhances safety and readiness for unforeseen circumstances at home.

List of Item for Pediatric Home Ventilation

ITEMS		ITEM CODE
Tracheostomy Tube	Shily	
	Neonatal 3.0	
	Neonatal 3.5	
	Pediatric 3.5	
	Neonatal 4.0	
	Pediatric 4.0	
	Neonatal 4.5	
	Pediatric 4.5	
	Pediatric 5.0	
	Pediatric 5.5	
	Portex	
	4.0	
	4.5	
	5.0	
Suction Items	Catheter size: Fr 6 (green)	
	Catheter size: Fr 8 (blue)	
	Catheter size: Fr 10 (black)	
	Catheter size: Fr 12 (white)	
Respiratory Items	Connection tube between suction machine and suction tube	
	Thermovent (white filter for tracheostomy)	
	Ped Nebulization kit	
	Oxygen tubing	
	Ventilator Nebulization kit	
	Ventilator tubing	
	Micro mount (green connection)	
	HME	
Feeding item	60ml syring	
	Tip 60ml syring	
	5 ml syring	
	2mlsyring	
	NGT tube size 6	
	NGT tube size 8	
	NGT tube size 10	
	Tegaderm	
	Plain IV set	
Tracheostomy care	Oral care	
	Tracheostomy care kit	
	PEDIATRIC TRACHEOSTOMY TIE	
	Gloves small	
	Gloves medium	
	Glover Large	
	Gauze	
Other equipment	Normal saline 0.95 Nacl	

Invasive home ventilation and home monitoring

Dr. Hussain Al Lawati
Pediatric Pulmonologist

- **Invasive home ventilation**

The mode of ventilation and settings vary according to the patient's medical condition, necessitating clinical judgment, discussion, and agreement between the primary physician and home ventilation physician. While there is no maximum limit for ventilatory settings in home ventilation, exhaustive consideration of all diagnostic and management strategies, particularly for those requiring high settings, is essential to discharge them on the lowest feasible setting.

Home ventilation settings are individualized based on various factors, with the decision on safe home settings determined by the home ventilation physician. In some instances, a gradual transition to home ventilator usage (e.g., few hours daily) may be preferable to a one-step transition from hospital ventilator.

Airway clearance techniques should be taught to children and caregivers as a preventative measure, particularly for those with evidence of impaired cough. Providing heated humidity through the ventilator circuit is essential, with heated humidity recommended over heat-humidity exchangers (HME) in children.

Before discharge, the child must be medically fit and stable, meeting all of the following criteria:

- Availability of all home equipment.
- Maintenance of stable and unadjusted ventilator settings for at least one week (e.g., Fio₂ less than 40%, PIP less than 28 cm H₂O, and PEEP equal or less than 10 cmH₂O).
- Maintenance of pCO₂ within acceptable safe limits on the home ventilator (with the level determined based on the clinical condition).
- Adequate nutritional intake to support expected growth and development.
- Well-trained family members proficient in tracheostomy care, emergency management, and equipment maintenance.

- **Home monitoring:**

The low minute ventilation alarm and disconnect alarm should be carefully configured to enhance the detection of circuit disconnection and decannulation. It is essential to optimize these settings to maximize safety. As a minimum standard, patients should be equipped with pulse oximetry for home monitoring, particularly during periods of sleep or when unobserved. This ensures continuous monitoring of oxygen saturation levels. It is important to verify and compare the pulse oximetry readings with those obtained from the hospital oximeter before discharge.

Relying solely on ventilator alarms may not be sufficient, especially in cases where the tracheostomy size is equal to or less than 5 mm. Accidental decannulation or ventilator disconnection incidents may not always trigger the low-pressure alarm on the ventilator. Therefore, pulse oximetry serves as an additional layer of safety to detect potential issues promptly.

Follow up

Dr. Saif Awlad Thani
Pediatric Intensivist

A comprehensive and documented follow-up plan should be established and clearly explained to the parents or caregivers. The frequency of follow-up appointments will be tailored to the child's individual condition and will gradually decrease as the parents gain confidence in managing their child's care at home.

Following home transition, the frequency of follow-up appointments will be as follows:

- Phone call: On the evening of discharge, 48 hours post-discharge, and 1-week post-discharge.
- Physical appointment: Scheduled after 4 weeks post-discharge.

- Subsequent appointments: Every 3 months thereafter.
- Elective admission: As determined by the physicians based on the child's ongoing medical needs.

This structured follow-up schedule ensures ongoing monitoring and support for both the child and their caregivers, promoting continuity of care and addressing any concerns or issues that may arise following home transition.

Nutritional support

Ameera Al Nassery
Dietician

Patients reliant on mechanical ventilation encompass a diverse group, ranging from those with chronic critical illness to individuals with chronic respiratory failure and various neuromuscular disorders. Despite this diversity, a prevalent characteristic among these patients is the heightened prevalence of malnutrition. This nutritional deficiency can significantly impact respiratory function, potentially resulting in prolonged or intensified reliance on mechanical ventilation.

Managing nutrition in this patient cohort presents challenges due to the heterogeneous nature of their metabolic needs. While many patients exhibit hypermetabolism, others may experience decreased metabolic requirements. Furthermore, respiratory dependency often coexists with additional comorbidities, further complicating nutritional management. The objectives of implementing a Nutrition Therapy Program (NTP) for ventilated patients include:

- Conducting a comprehensive assessment of the patient's nutritional status.
- Precisely determining energy requirements to prevent both over- and underfeeding.
- Developing a well-structured nutrition plan tailored to the individual patient's specific needs, with ongoing close monitoring.
- Ensuring adequate protein intake, crucial for preserving and restoring muscle mass and strength.
- Administering nutrition via oral intake or through enteral/parenteral nutrition services, as appropriate.
- Addressing dysphagia-associated nutritional deficiencies, which are common yet often undiagnosed early, necessitating collaboration with speech and swallowing therapists for effective management.

Feeding with tracheostomy

Dr. Amir Al Tubi
Speech pathologist

Speech-Language Pathologists (SLPs) assess the suitability of initiating oral feeding once approval is granted by the medical team, confirming the patient's medical readiness for oral trials. Initially, a bedside assessment may be conducted, followed by a more comprehensive objective evaluation of the patient's swallowing function, as needed, using methods such as Video Fluoroscopic Swallowing Study (VFSS), Fiberoptic Evaluation of Swallowing (FEES), or a blue dye test, if feasible, to detect any signs of aspiration during feeding.

Feeding Procedure:

For patients with an inflated cuffed tracheostomy, it is recommended to deflate the cuff before feeding, as an inflated cuff may not entirely prevent aspirated materials and excessive secretions. In fact, the cuff may act as a reservoir for aspirated and pooled secretions, increasing the risk of aspiration pneumonia. Adequate oral care should be administered before and after feeding, as patients with a tracheostomy may experience altered upper airway function, potentially leading to bacterial colonization and an increased risk of pneumonia.

General guidelines for all patients include:

- Suctioning before oral feeding or via the tracheostomy tube, if applicable.
- Positioning the patient in an upright position with the head slightly flexed or providing adequate head support.
- Continuous monitoring of oxygen saturation and heart rate throughout the feeding process.
- If the patient has been fitted with a speaking valve, it should be placed before feeding, as the restoration of subglottic pressure may help reduce the risk of aspiration.

Communication with tracheostomy (speech valve)

Dr. Aamir Al Toubi
Speech pathologist

Patients with a cuffed tracheostomy often experience speech loss, which can cause distress for both the patient and caregivers. This loss of communication ability may lead to fear, depression, or feelings of isolation. To address this issue and improve communication, the following steps are recommended for new patients with tracheostomies who are suitable for using communication aids:

- Referral to Speech-Language Pathologist (SLP) for assessment of suitability of communication aids.
- Initial assessment by the SLP, evaluating the patient's ability to see, hear, touch, write, understand, or use facial expressions.
- Communication with the patient using the recommended method determined by the SLP.
- Assessment of established tracheostomies for the use of speaking valves if appropriate.
- Evaluation by the SLP to determine suitability for using a speaking valve, which allows air to pass into the tracheostomy during inhalation but not out during exhalation, thereby enabling vocalization.

Criteria for suitability to trial speaking valve include:

- Ability to tolerate cuff deflation for patients with cuffed tracheostomies.
- Absence of airway obstruction.
- Stable medical/pulmonary status.
- Ability to manage own secretions.
- Appropriate tracheostomy tube size.
- Appropriate ventilator settings.

Procedure for all staff for commencement of speaking valve use:

- Position the patient and suction through tracheostomy and oral.
- Gradually deflate the cuff for patients with cuffed tracheostomies, observing reactions such as coughing, gagging, swallowing, and grimacing.
- Inform the patient about the impending placement of the speaking valve.
- Occlude the tracheostomy for 1-5 seconds to prepare the patient for the change in respiration when the speaking valve is placed.
- If the patient tolerates well, place the speaking valve securely but not too tight at the end of the tracheostomy tube with a gentle twist.
- Stabilize the tracheostomy when placing or removing the speech valve to avoid irritation and coughing.
- Adjust the ventilator settings and alarms with the respiratory therapist and pediatric intensivist to ensure appropriate ventilation while the speech valve is in place.
- Monitor vital signs and remove the valve if the patient experiences difficulty breathing, distress, or a drop in oxygen saturation.
- Remove the speaking valve when the patient is asleep, as patients should not sleep with the valve in place until a full trial is completed.

Options for communication for ventilated patients include cuff deflation, speech valve, electro larynx, and talking tracheostomy tube. The speaking valve should be cleaned with a mixture of water and a sterilizing tablet, avoiding the use of hot water, peroxide, bleach, vinegar, alcohol, brushes, or Q-tips.



Home environment

Nuha Al Shaaili
Occupation therapist

Families are provided with detailed instructions on how to prepare the home environment to ensure the safety of the child. With multiple equipment in the house, it is crucial to ensure electrical safety and eliminate environmental hazards. Here are some guidelines provided to the parents:

1. Ensure good lighting and enough space in the room where the child sleeps so that all home devices can be placed appropriately and caregivers can provide care easily. Organize the equipment well in the room for easy access.
2. Keep the ventilator out of direct sunlight and away from windows.
3. Do not place the ventilator or oxygen near any fire hazards, such as frankincense burners, stoves, open flames, heaters, cigarettes, or lighters, to prevent overheating or fire hazards (oxygen is flammable).
4. Do not block the inlet filter on the ventilator or the nebulizer, and replace damp filters promptly.
5. Ensure space between the ventilator and external battery to prevent overheating.
6. Place the humidifier below the head of the child's bed.
7. Keep emergency equipment and supplies readily available.
8. Ensure a working phone is accessible for emergency calls.
9. Prepare an emergency travel bag for immediate use.
10. Secure oxygen tanks to prevent tipping and potential fire hazards.
11. Organize drawers and storage areas for easy access to medications and treatment supplies, ensuring good lighting for accurate dosages.
12. Circle expiration dates on medications and reorder refills at least one week ahead.
13. Store monthly supplies separately from daily supplies for easy access.
14. Maintain cleanliness to protect the child from infections, using good hand hygiene and alcohol-based gel hand sanitizers. Avoid contact with the child if family members are sick.

Day care visit

Dr. Saif Awlad Thani
Pediatric Intensivist

Home ventilated patients attend daycare for follow-up every 3 to 6 months, where the following procedures are carried out:

- Tracheostomy change, preferably performed by a parent under supervision.
- Ventilator circuit change if not already done by the parent at home. Circuit changes are typically conducted monthly, and parents are trained by respiratory therapists on how to perform this task at home. They are also provided with the necessary circuit by the respiratory therapist.
- Ventilator checks conducted by respiratory therapists to ensure proper functioning.
- Review of the patient's clinical condition and addressing any concerns raised by the parents.
- Laboratory investigations are performed if deemed necessary.
- Review by other subspecialties as required.

Home Simulation Day

Dr. Saif Awlad Thani
Pediatric Intensivist

A period of "care-by-parent" is the final step before discharge, during which the child remains an inpatient but is managed by caregivers with support as needed from nursing staff or respiratory therapists.

Home Simulation Day occurs when a patient planned for discharge with a tracheostomy and/or ventilation is moved from the intensive care or high dependency unit to the general ward one or more days before going home. During this time, the family provides full care for the child as they would at home, while doctors and nurses monitor and supervise to ensure the family's competency and safety before discharge.

During Home Simulation Day, the child is placed in a room or bed on the ward, and at least two caregivers, ideally both parents, are present to provide support. The discharge planner arranges appropriate space for both parents, the patient, and equipment in advance. If both parents cannot be present simultaneously due to ward circumstances, one parent must attend at a time.

All home equipment, including the ventilator, suction machine, oxygen concentrator, and home saturation monitor must be used. Continuous hospital monitoring is not necessary, but routine vital sign checks are conducted per ward policy.

Home Simulation Day should only proceed after caregivers pass all competency checklists assessed by the pediatric home ventilation team and when the patient is deemed ready for home ventilation from a medical standpoint. All home equipment must be operational, and the family must feel confident in using it.

This simulation allows the family to experience home ventilation care independently while still surrounded by support. It boosts their confidence, reduces anxiety, and prepares them for home care. Additionally, it allows the medical team to assess the family's readiness, skills, reactions, competency, and independence, identifying areas needing further training and addressing any concerns before discharge. It also provides an opportunity for the family to ask questions or discuss concerns that may not have arisen when care was shared between them and the healthcare team in the intensive or high dependency unit.

Downsizing and Decannulation

Dr. Hussain Al Lawati
Pediatric Pulmonologist

Decannulation:

Predictors for successful decannulation include:

- Patient is in good general health.
- Underlying reason for requiring the tracheostomy has resolved.
- No barriers to decannulation exist, such as:
 - Inadequate vocal cord function.
 - Increased secretion.
 - Recent/current infections.
 - Optimal nutritional status.

Relative contraindications to weaning are:

- Documented aspiration.
- Tracheal esophageal fistula.
- Bleeding from tracheostomy site.

Decannulation criteria include:

- Patient can spontaneously maintain ventilation without ventilator support for at least 2 months.
- Movement of air around the tracheostomy tube occurs with tracheostomy tube occlusion. This may require downsizing the tube to a smaller size.
- Patient can swallow adequately (SLP +/- OT assessment).
- Patient exhibits a strong cough either spontaneously or with suctioning.
- Adequate oxygenation on less than 40% FiO₂.

Steps needed before decannulation:

- Upper airway assessment to rule out any upper airway obstructive pathology like subglottic stenosis, bilateral vocal cord palsy, suprastomal granuloma, etc.
- The tracheostomy tube should be capped all day, and the cap removed at night for several weeks to determine whether the cap can remain when the child has a URI.
- Prior to decannulation, children with a tracheostomy should undergo a daytime capping trial (if they are older than 2 years and if the tracheostomy does not occupy so much of the trachea so as to preclude capping).
- If they pass the daytime capping trial, the next step is either a capped sleep study or a nighttime capping trial while hospitalized and being observed.
- If the patient passes all above requirements, decannulation is done in the hospital and the patient is kept for at least 48 hours to monitor day and night saturation and breathing pattern.

Admission for re-evaluation

Dr. Ali Al Mawali
Head of general pediatric team

The Pediatric Home Ventilation Team serves as a consulting entity and does not possess admission privileges. In cases where Home Ventilated patients require elective admission for purposes such as evaluation of ventilation requirements, possible weaning, decannulation, or family re-teaching, admission is facilitated under the general pediatric team or pulmonology team after consultation with the team consultant. Both the Pediatric Home Ventilation Team and the admitting team collaborate to guide the management plan effectively. If elective admission is necessary for evaluation by any other team unrelated to home ventilation, the patient is admitted by the relevant team, and the Pediatric Home Ventilation Team is consulted specifically for matters concerning tracheostomy or ventilation.

Admission to the ward and monitoring policy

Asma Al Naabi, Dr. Aza Al Sawafi, Dr. Hussain Al lawati, Dr. Saif Awlad Thani

Patients on home ventilation through a tracheostomy tube require meticulous care and observation when admitted to general wards. This policy outlines the necessary measures and responsibilities to ensure their safety and well-being:

Mandatory Safety Requirements:

- The accompanying family member or caregiver must be familiar with routine care.
- Ventilation settings must mirror home settings, with no alterations without notifying the home ventilation physician or team consultant.
- Continuous monitoring, especially during sleep, is essential for all home ventilated patients.
- Respiratory therapists must see patients three times daily during hospital stays.
- All alarm settings must remain activated.
- Ventilator screens must be locked.
- Emergency equipment, including appropriately sized tracheostomy tubes, an Ambu bag, suction tube, suction machine, and tracheostomy ties, must be bedside.

Role and responsibility of the admitting team:

For patients whose caregiver has received training at the Royal Hospital:

- Refer to the pediatric home ventilation team within 24 hours of admission.
- Refer to respiratory therapists upon admission.
- Arrange referrals to subspecialty teams as needed.
- Address patient requirements, including appointments and tracheostomy changes.

For patients whose caregiver has been trained elsewhere:

- Follow the same protocol as above.
- Additionally, refer to patient flow services, social worker, and speech therapist.

Role and Responsibility of Nurses:

Ward Nurse (In-Charge Nurse):

- Assess patient and caregiver needs and coordinate with other teams.
- Ensure emergency equipment is bedside.

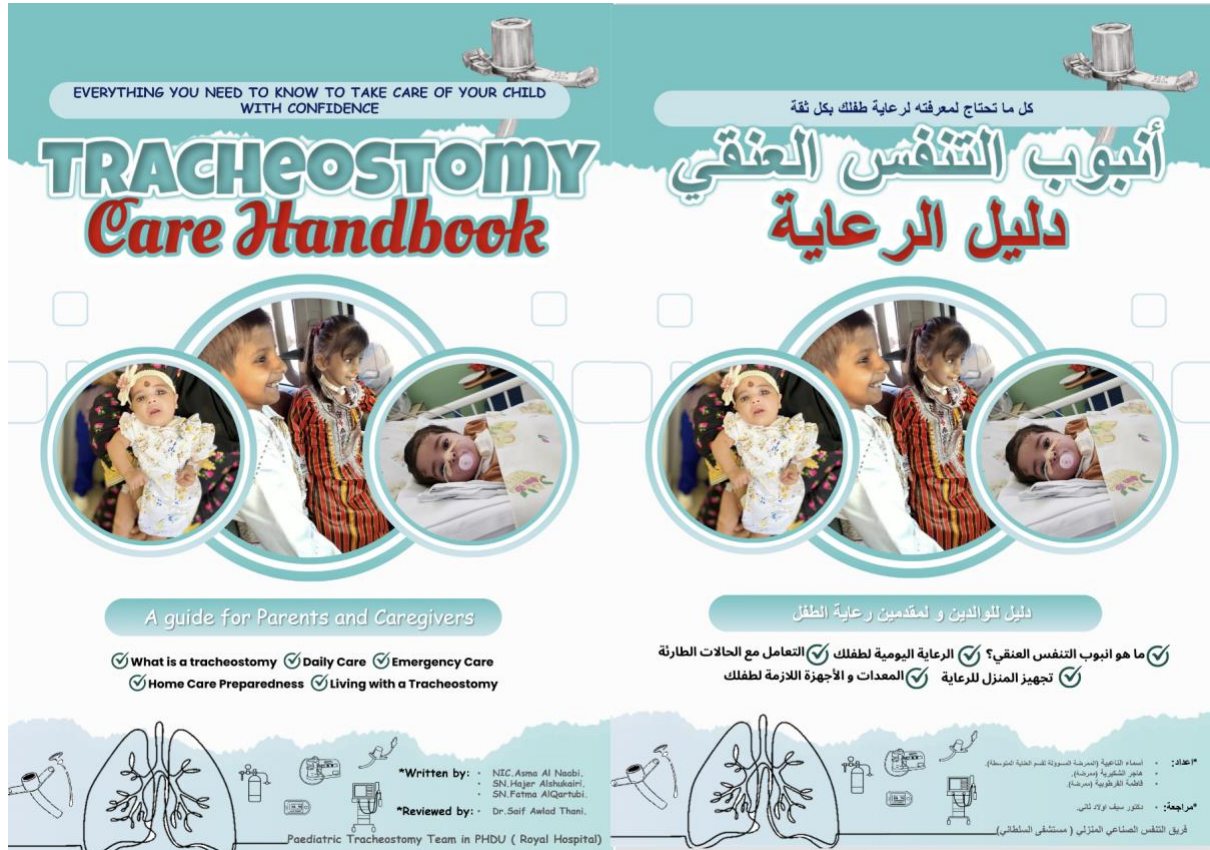
Bedside Nurse:

- Ensure continuous patient monitoring.
- Confirm emergency equipment accessibility.
- Conduct two-hourly purposive rounds.
- Assist and assess the caregiver.
- Alert doctors to any changes or family concerns.
- Nurses must provide care from 22:00 to 05:00 to allow caregivers adequate rest.

Role and Responsibility of the Home Ventilation Team:

- For patients discharged from the Royal Hospital:
 - Assess tracheostomy and ventilation requirements and address concerns.
 - Collaborate with team members for optimal care.

- For new patients:
 - Review within 24 hours of referral.
 - Assess requirements and collaborate with respiratory therapists.
 - Evaluate caregiver skills and provide necessary teaching.
 - Collaborate with team members for comprehensive care.



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